

Morphological characterization of sacrificial layers of porous silicon for Photovoltaic application

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Abstract. We investigate the application of porous silicon (PS) in the field of the photovoltaic. In first step we realise a double porous silicon layer by electrochemical anodisation using two different current densities. The low current leads to a low porosity and during annealing, the recrystallisation of this layer allows epitaxial growth. The second current leads to a high porosity which permits the transfer onto a low cost substrate. During high temperature onto hydrogenation treatments of to passivate the structure and epitaxy in liquid phase, porous silicon is recrystallized partially. In this work, a characterization by scanning electron microscopy informs us about the morphology of these porous layers.

Introduction

The crystalline silicon cells with thin layer are of great interest in the photovoltaic field. However, two problems are to be solved: to find the good substrate with low costs, having the same thermal dilation coefficient as silicon, as well as the adequate technique to deposit single-crystal silicon there. Like substrates with low costs, ceramics was used in several research tasks [1, 2]. However the direct realization such cells, requires a plug layer having to play the part of support for the epitaxy layer. Requiring a great technological control and the realization of adequate plugs layers, this technique industrially remains not very gravitational. A good candidate for an improvement of the quality ratio/price of the solar cells of terrestrial use is porous silicon (PS). Indeed, the realization of porous silicon require an anodization of single-crystal silicon in hydrofluoric acid and methanol. The growth by epitaxy in liquid phase (EPL) on the porous one was used for the first time in 1978 [3], it remained however to find a means to separate the epitaxy layer in order to stick it on a substrate to low costs like ceramics. This problem was solved [4] by using a porous layer gradual (low porosity over and strong in lower part) comprising a sacrificial layer, located in its medium as proposed by Bergmann [5] and Tayanaka [6]. In the realization of the solar cells, PS undergoes the following stage of heat treatments under various atmospheres:

- annealing under hydrogen at 450°C during 15min to desorb porous silicon then and folowed by 950°C – 1050°C to simulate the stage of epitaxy in liquid phase,
- annealed at 1000°C under nitrogen gas to simulate the development of the junction then 700°C/30s under air to obtain a good contacts.

In this study we are interested in the evolution of the microstructure of porous silicon layer and porous layer sacrificial during the heat treatment.

Experimental procedures

Electrochemical process

A p-type substrate (100) with a resistivity 0.01-0.02 Ω .cm was used in this work. The porous structures are obtained by anodic treatment of substrates of Si. This electrochemical attack is carried out in a HF/C₂H₅OH solution (1/1 by vol) using two current densities 5 and 75 mA/cm² at different etching time 2, 3, 10 and 20 min. Using two constant current densities allows to carry out different porosities. The low density leads to a low porosity and conversely, while the etching time

determines the thickness of the desired porous layer. The ethanol limits the accumulation of hydrogen bubbles formed during the reaction of anodization, thus increasing the homogeneity of the layers.

An electrochemical etching is carried onto p-Si, with times and densities of variable current so as to obtain gradual layers being formed from surface (in-depth porosity) thus forming the sacrificial layer of porous silicon.

All the samples are immersed in the deionized water and dried with nitrogen gas. The various samples prepared in this study are described in table 1.

Heat treatment

The prepared samples are annealing at 450°C during 15 or 30 min under hydrogen and were heated at 950°C under H₂ then with 1050°C under neutral atmosphere in order to compare the effect of the atmosphere in annealing. Having undergone this treatment, the same samples undergo an annealing at 1000°C during 1 hour under N₂ in order to simulate the stage of diffusion of doping agents to form the junction n⁺p and finally a thermal 700°C treatment under air in order to simulate the stage of realization of the contacts. These various heat treatments can influence the microstructure of the porous layer.

After annealing, the samples were the subject of microstructural analysis by using the scanning electron microscopy (SEM) summarized in table 1.

Samples Si (100) type p	1HF (48%)/1C ₂ H ₅ OH		
	Current density J(mA/cm²)	Etching time	Heat treatment (atmosphere temperature time)
A	5	10	H ₂ -950°C-2h
B	75	10	H ₂ -950°C-2h
C0	5 and 75	2 and 2	(1) H ₂ -950°C-2h, (2) N ₂ -1000°C-1h, (3) Air-700°C-30s
C1	5 and 75	1 and 1	(1) H ₂ -950°C-2h, (2) N ₂ -1000°C-1h, (3) Air-700°C-30s
C2	5 and 75	1 and 5	(1) H ₂ -950°C-2h, (2) N ₂ -1000°C-1h, (3) Air-700°C-30s

Table 1: Summarized various conditions of preparation of the samples and their heat treatments .

Results and discussion

Scanning Electron Microscopy

SEM observation reveals that the thickness of the porous layers depends on the current density value and on the etching time respectively. On micrographies obtained, it appears in an obvious way that the texture of the porous layer depends greatly on the current density and its thickness depends on the etching time.

The cross section of the porous layer shows an anisotropic etching with a preferential orientation. The pores are consisted of channels whose average diameter varies from 3 to 1 nm that one cannot observe using SEM. These channels are propagated in all the thickness of the layer and can thus reach several microns length.

The surface morphology of the sample shows that after thermal annealing, the porous layers are recrystallized and pores layers of important porosity increase. The morphological analysis of porous silicon was carried out in the case of three samples A, B and C summarized in table 1:

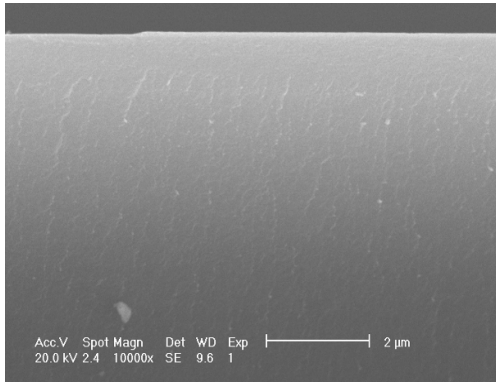


Fig. 1: Cross sectional SEM image of porous structure of the A sample before annealing.

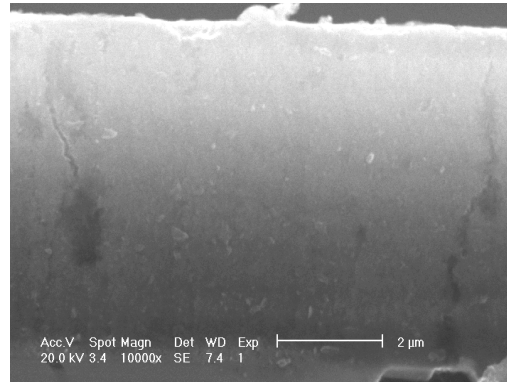


Fig. 2: Cross sectional SEM image of porous structure of the A sample after annealing by H_2

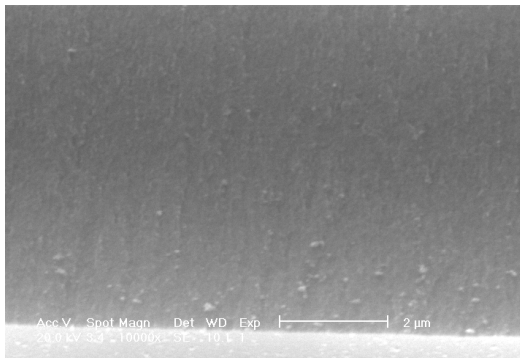


Fig. 3: Cross sectional SEM image of B sample before annealing.

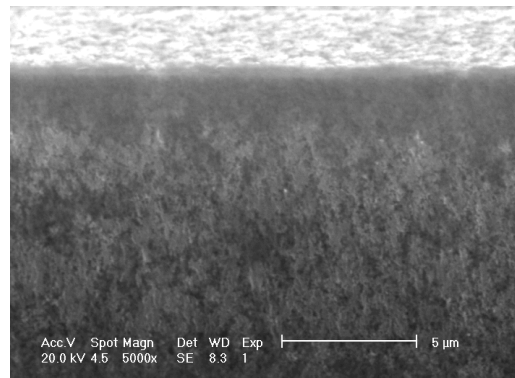


Fig. 4: Cross sectional SEM image of B sample after annealing by H_2 .

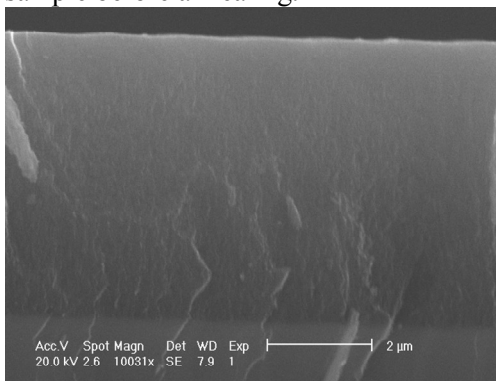


Fig. 5: Cross sectional SEM image of C_1 sample before annealing.

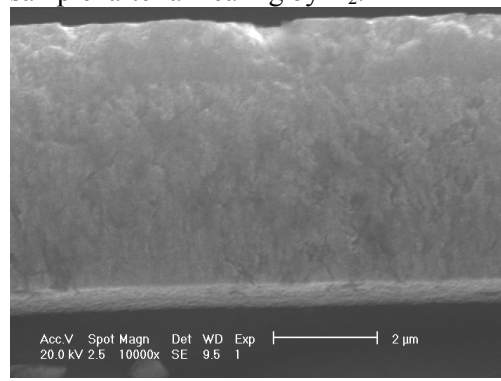


Fig. 6: Cross sectional SEM image of C_1 sample after annealing by H_2 .

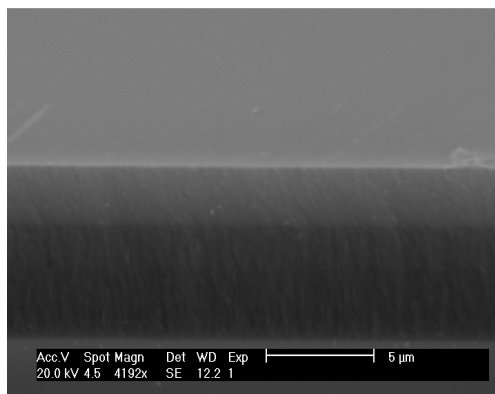


Fig. 7: Cross SEM image of C₂ sample before annealing.

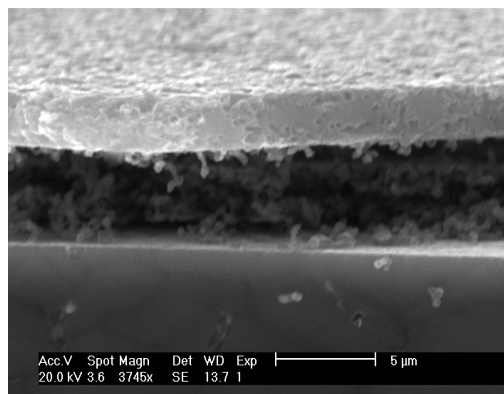


Fig. 8: Cross SEM image of C₂ sample after annealing by H₂ at 1050°C

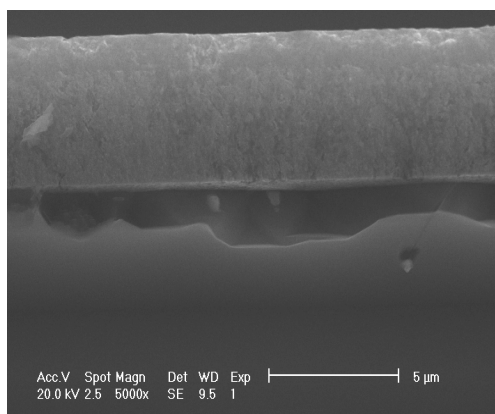


Fig. 9: Cross sectional SEM image of C₁ sample after annealing by hydrogen.

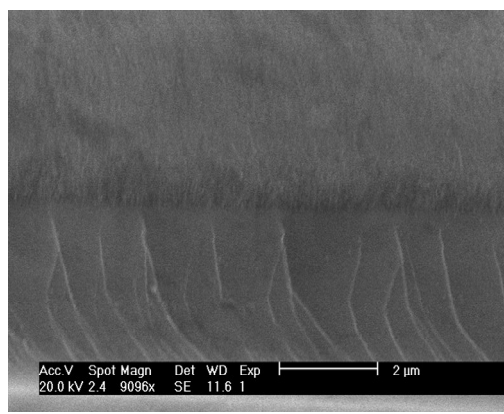


Fig. 10: C₁ Interface substrate/ PS after annealing by hydrogen, nitrogen then air.

The samples A (Fig.1, Fig. 2) and B(Fig. 3, Fig. 4) seen on profile, showed that after heat treatment, the structure of PS presents a rearrangement of the structure, a coalescence of the pores close to surface indicating a filling of the pores. This coalescence can be explained by the diffusion of the atoms of Si on the surface where the energy of surface is weaker. If a layer of SiO₂ is present on the walls of the pores, the diffusion of the atoms is blocked [8]. With regard to the gradual layers which undergo all annealings necessary for the growth of silicon and the development of the solar cell, we did not observe premature setback of porous silicon (Figs. 9-10). Nevertheless after annealing under hydrogen, the interface substrate/PS is attacked by places (cf Fig 6, Fig.8) but this phenomenon is not visible anymore after cleavage of the samples (Fig. 10). It will be able thus, perhaps, exploited for the unhooking of the layers. After annealing under hydrogen, the layers of porous silicon do not seem any more to undergo a major modification of the structure during following annealings, under nitrogen and air (Fig. 10).

Conclusion

An electrochemical etching was carried out on (100) on p-type silicon for two current densities and etching times. It has been shown that high temperature treatment of the sample in H₂ leads to a partial cristallisation of the porous layers. The thickness of the porous layers varies with the temperature treatment. Finally, it has suggested that oxidation in air of PS samples affects the microstructure.

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